מכון טכנולוגי חולון
Holon Institute of Technology

PCB Design

קורס מס': 50177

Schematic Simulation

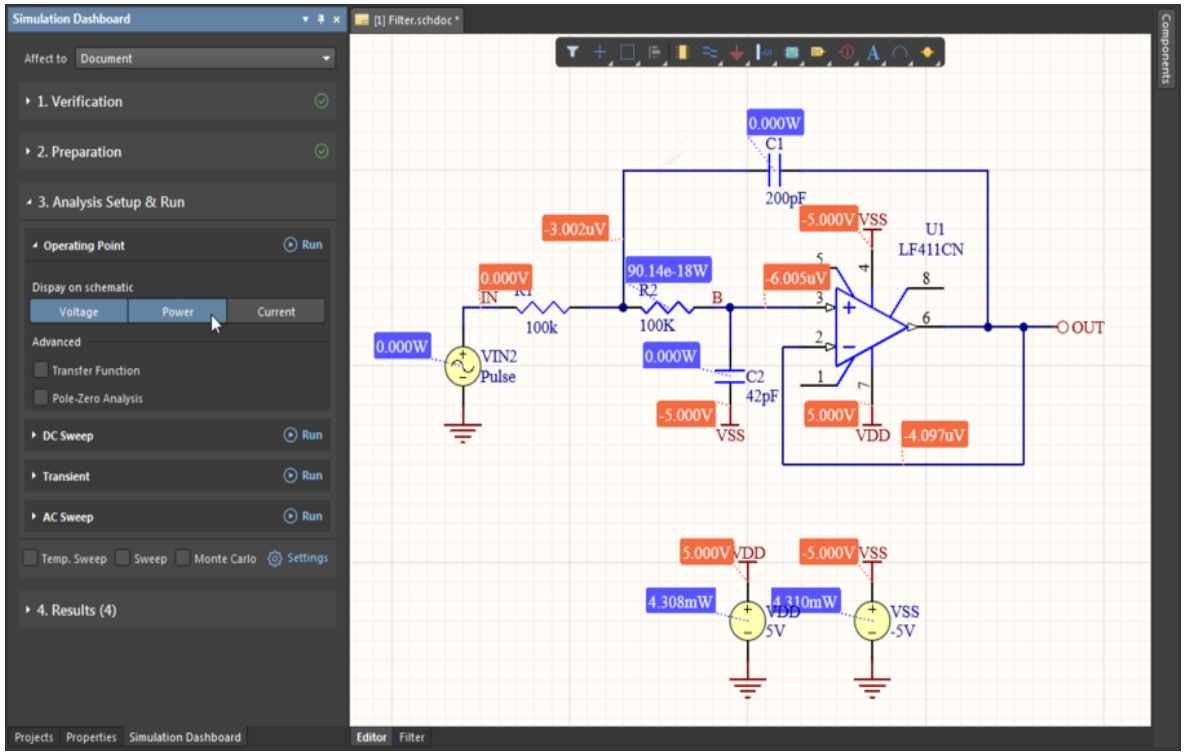
מרצה: ד"ר אביחי אהרון

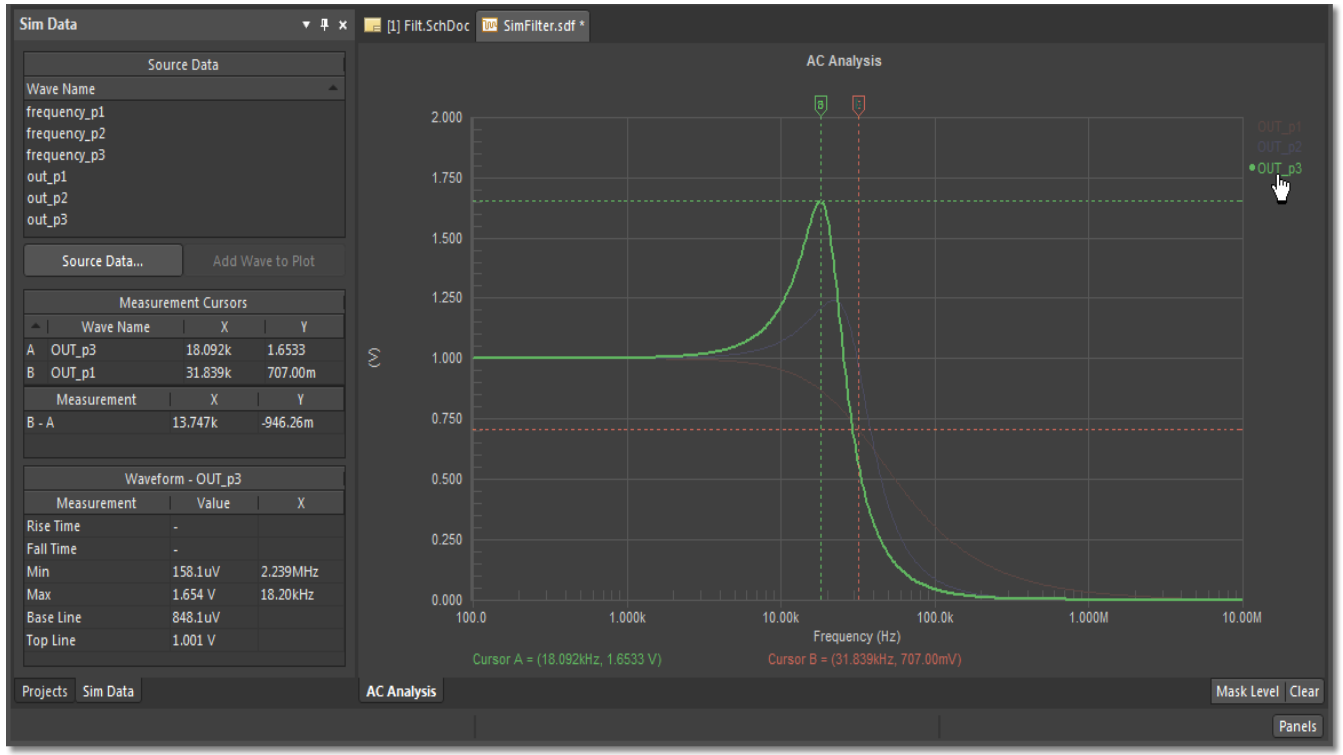
Schematic Simulation

SPICE

Introduction

SPICE ("**Simulation Program with Integrated Circuit Emphasis**")
SPICE is a general-purpose, analog electronic circuit simulator. It is a program used in integrated circuit and board-level design to check the integrity of circuit designs and to predict circuit behavior.





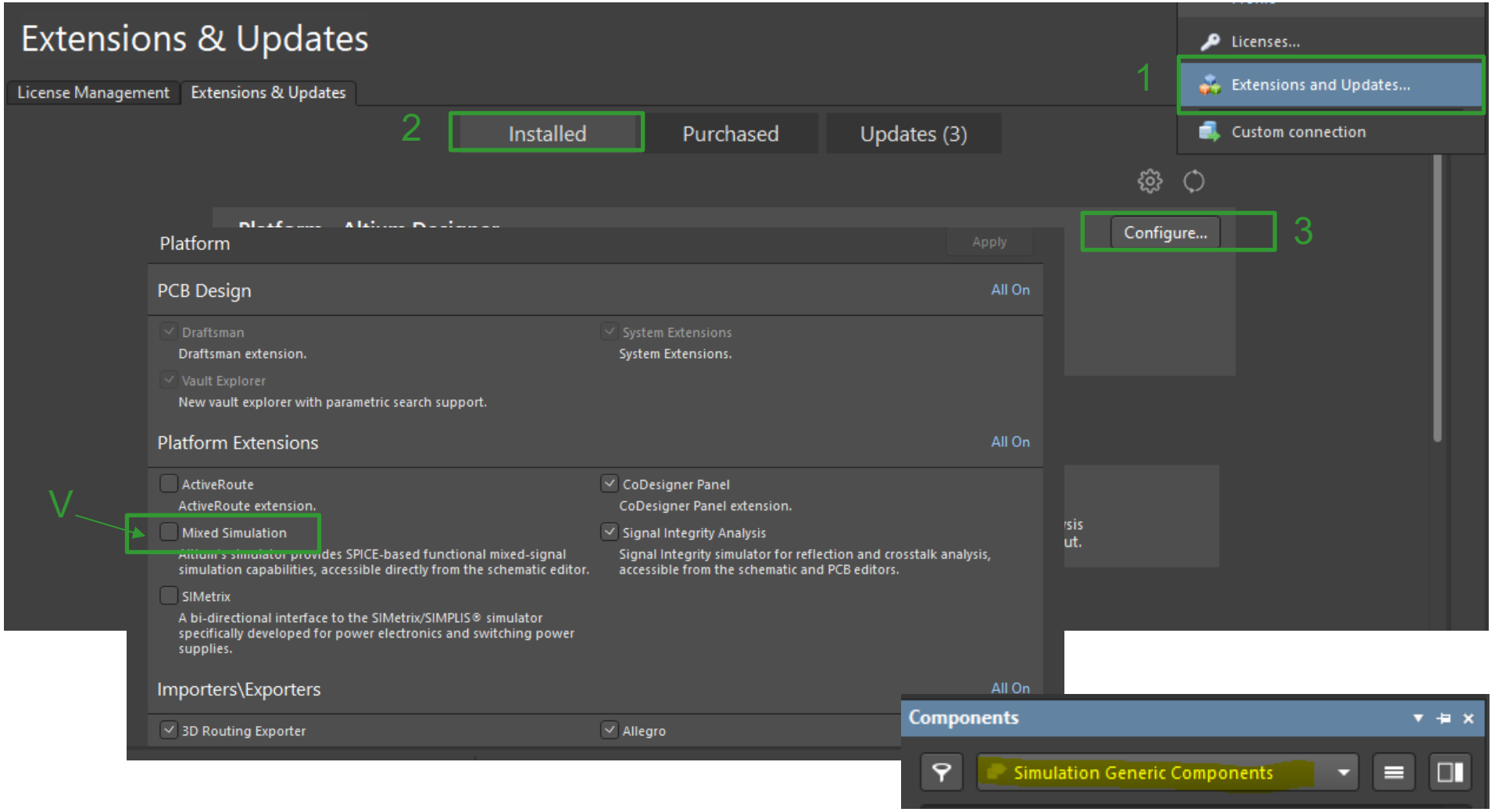
Enabling the SPICE

Drawing the Schematic

Simulation dashboard panel

Exercise

Enabling the extension "Mixed Simulation" in the first time.



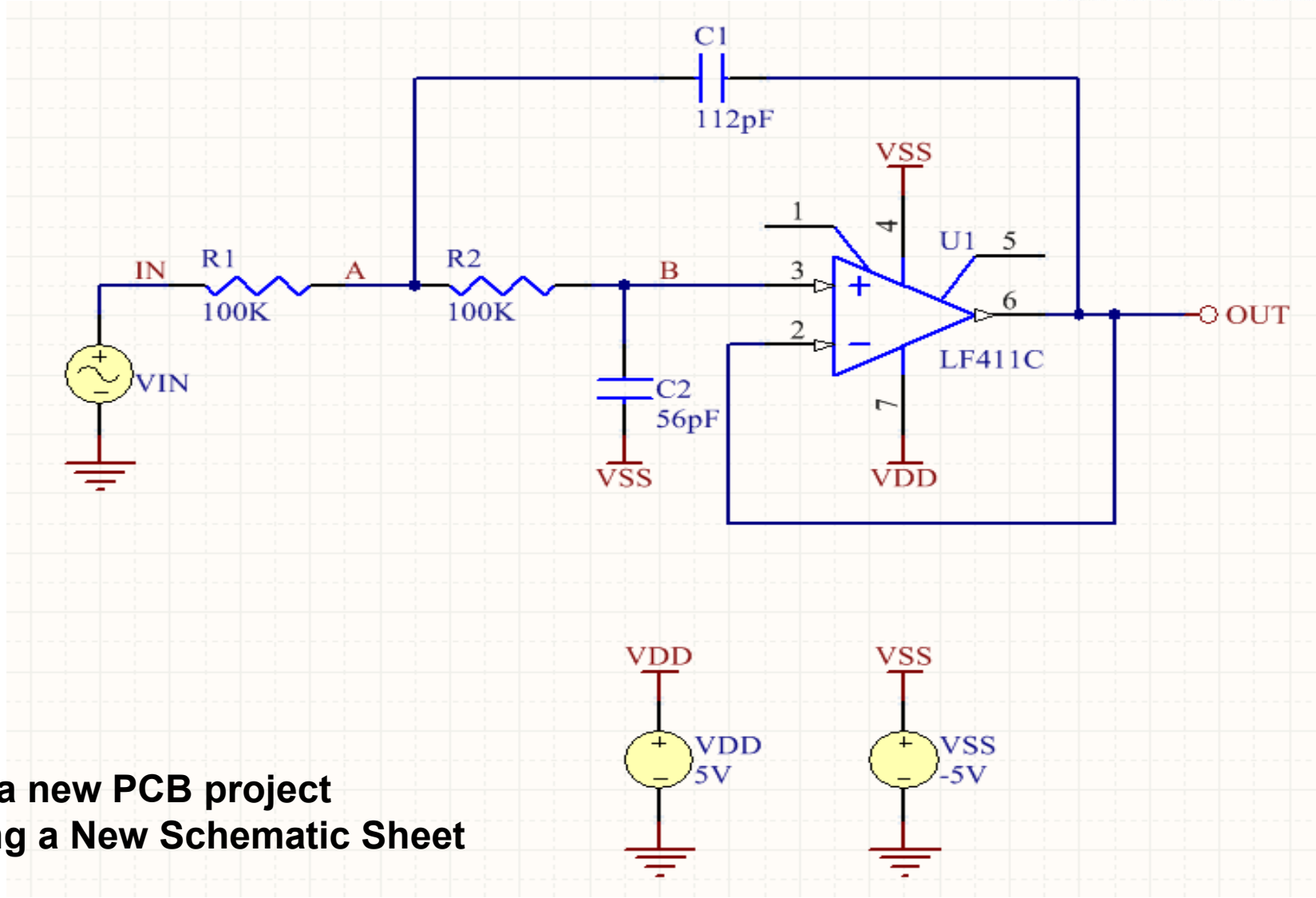
Schematic Simulation

Enabling the SPICE

Drawing the Schematic

Simulation dashboard panel

Exercise



1. create a new PCB project
2. Creating a New Schematic Sheet

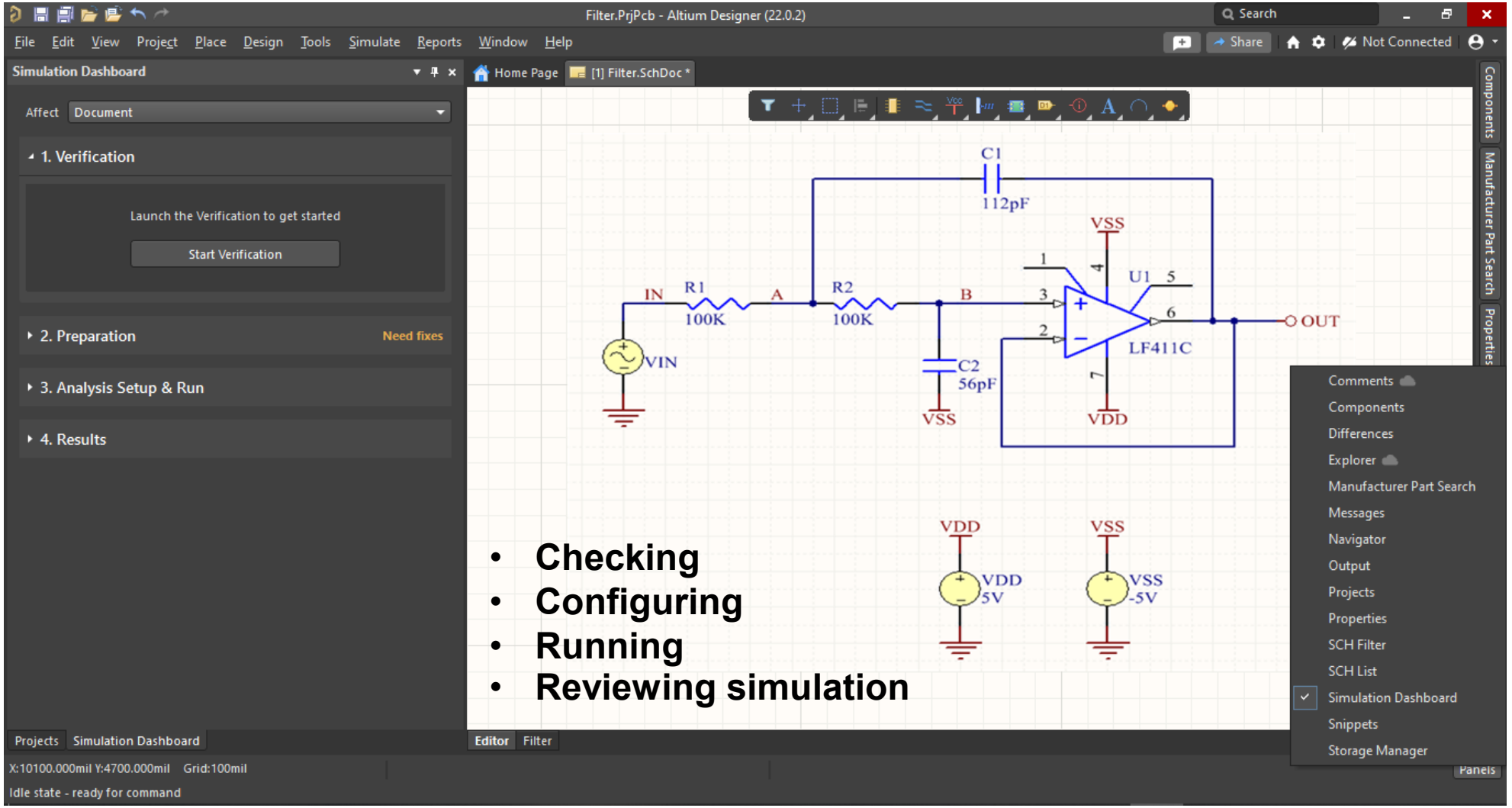
Schematic Simulation

Enabling the SPICE

Drawing the Schematic

Simulation dashboard panel

Exercise



- Checking
- Configuring
- Running
- Reviewing simulation

Schematic Simulation

Enabling the SPICE

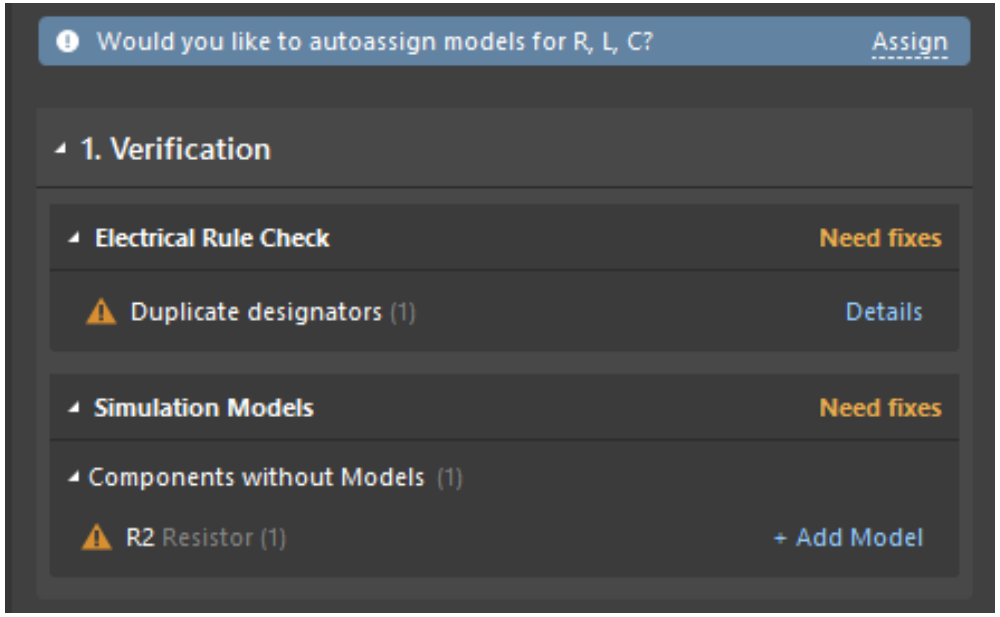
Drawing the Schematic

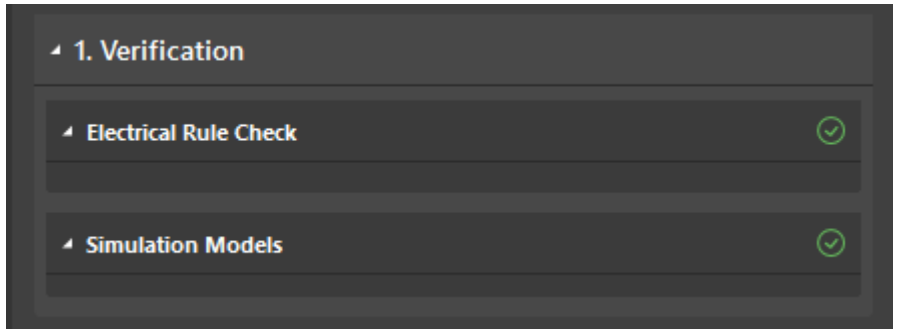
Simulation dashboard panel

Exercise

Verification:

- confirm that electrical rules checks are passed
- all components have simulation models.





Missing models or simulation source issues are flagged, with links to help resolve them.

Enabling the SPICE

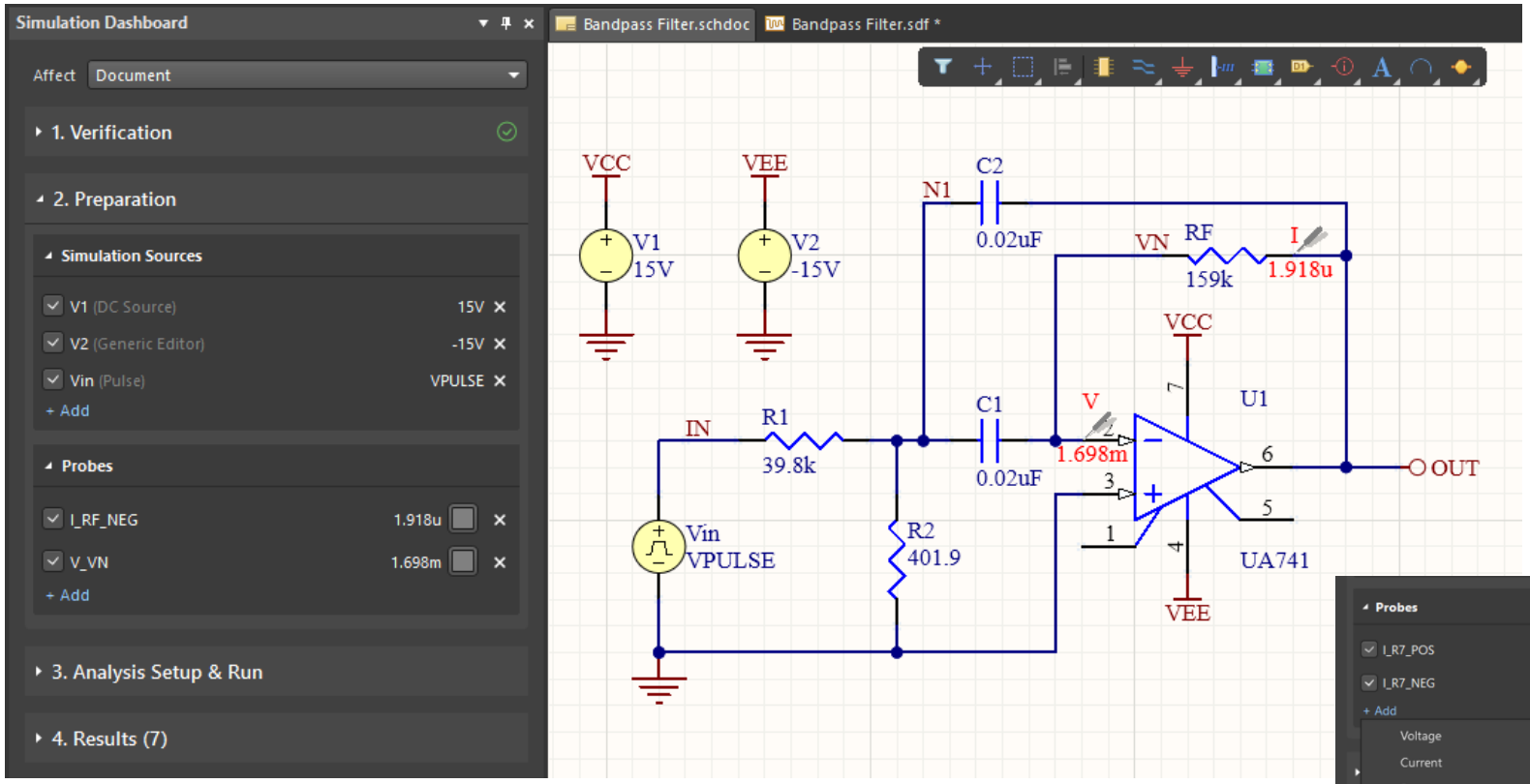
Drawing the Schematic

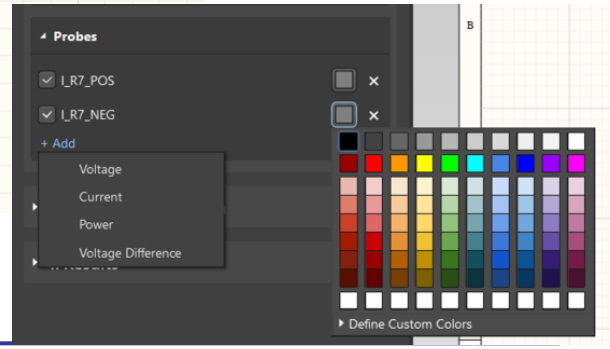
Simulation dashboard panel

Exercise

Preparation

Simulation Sources and Probes are listed (click to jump to a source/probe), and additional Sources or Probes can be added.





All signal sources and probes that are placed in the electrical circuit are shown in the Preparation area.

Enabling the SPICE

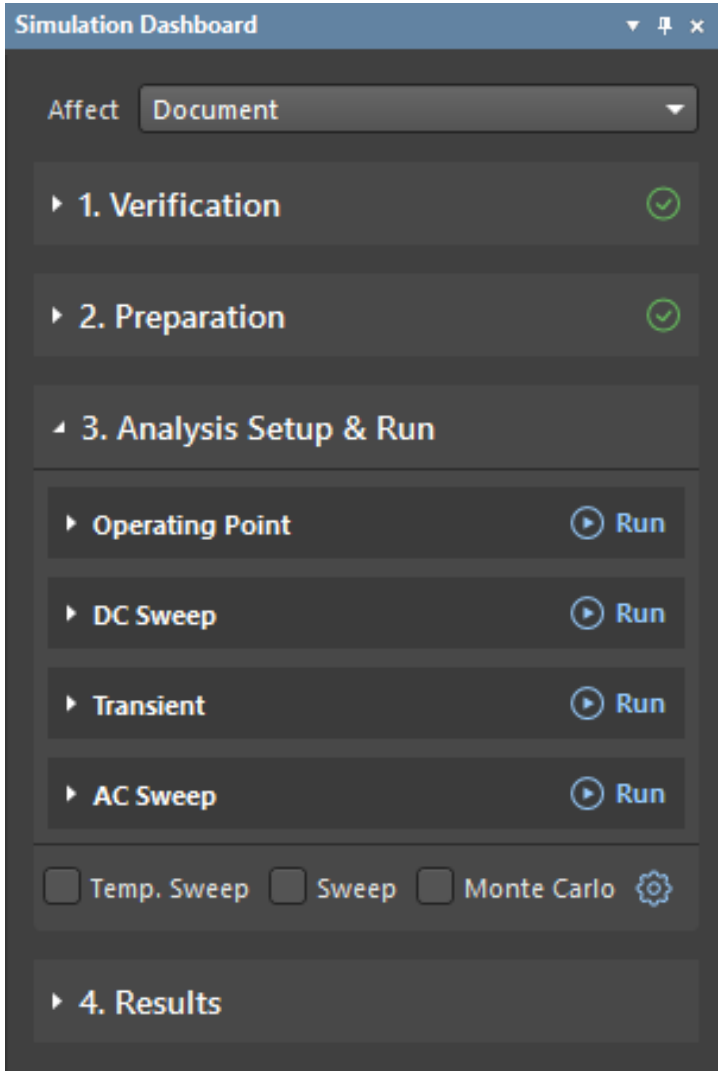
Drawing the Schematic

Simulation dashboard panel

Exercise

Analysis Setup & Run

Quickly configure the required analysis type and the Output Expressions to be plotted, then run the simulation.



Enabling the SPICE

Drawing the Schematic

Simulation dashboard panel

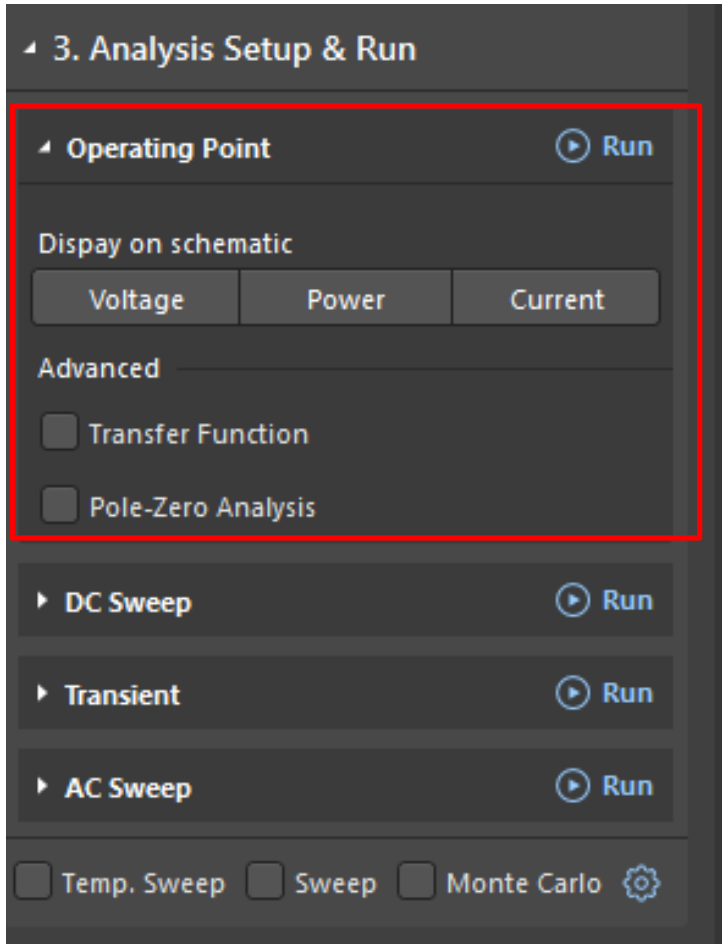
Exercise

Analysis Setup & Run

Quickly configure the required analysis type and the Output Expressions to be plotted, then run the simulation.

Operating Point - determines the dc operating point of a circuit, with inductors shorted and capacitors opened.

- **Poles and Zeros of transfer function** - used to determine the stability of a single input, single output linear system, by calculating the poles and/or zeros in the small-signal ac transfer function for the circuit. The dc operating point of the circuit is found and then linearized, small-signal models for all non-linear devices in the circuit are determined. This circuit is then used to find the poles and zeros that satisfy the nominated transfer function.
- **DC Transfer function** - (DC small signal analysis) calculates the DC input resistance, DC output resistance and DC gain, at each voltage node in the circuit.



Enabling the SPICE

Drawing the Schematic

Simulation dashboard panel

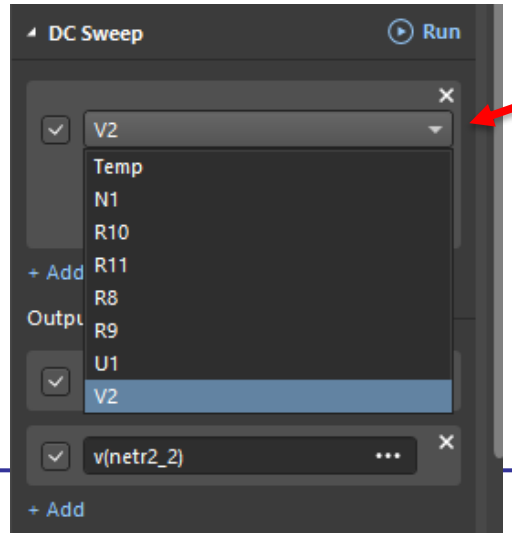
Exercise

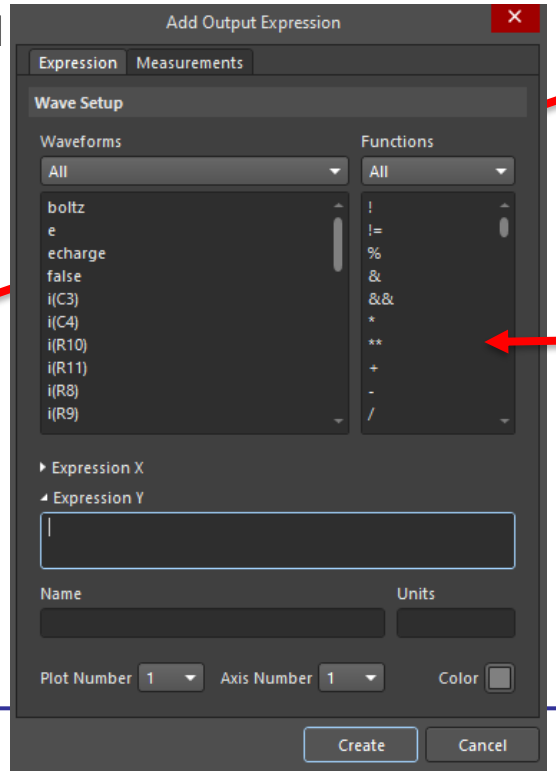
Analysis Setup & Run

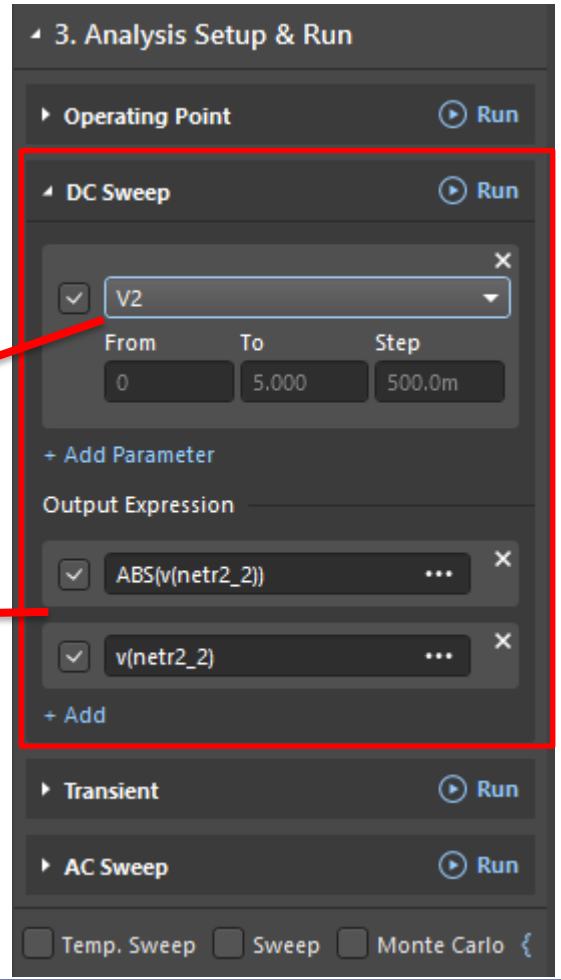
Quickly configure the required analysis type and the Output Expressions to be plotted, then run the simulation.

DC sweep - generates output like that of a curve tracer. Used to sweep variables, including: temperature, voltages and currents from sources, resistances and conductivity.

Assumptions: AC sources are zeroed out, capacitors are open, inductors are shorted, digital components are treated as a large resistor to ground.







Enabling the SPICE

Drawing the Schematic

Simulation dashboard panel

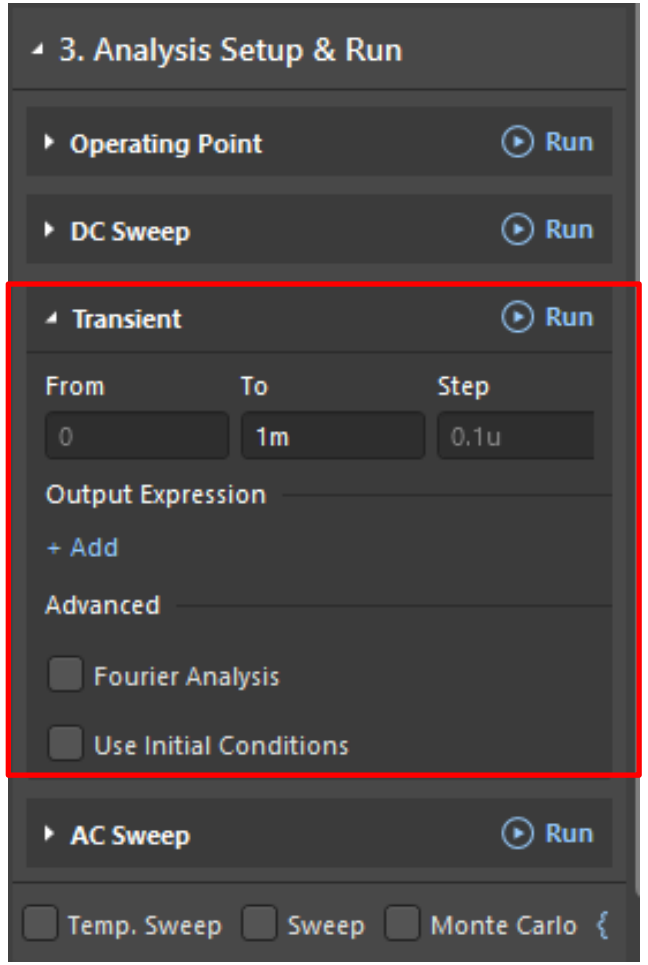
Exercise

Analysis Setup & Run

Quickly configure the required analysis type and the Output Expressions to be plotted, then run the simulation.

Transient analysis - generates output similar to that normally shown on an oscilloscope, computing the transient output variables (voltage or current) as a function of time, over the user-specified time interval. An Operating Point analysis is automatically performed prior to a Transient analysis to determine the DC bias of the circuit, unless the Use Initial Conditions parameter is enabled.

- **Fourier analysis** - Fourier analysis of a design is based on the last cycle of transient data captured during a Transient analysis. For example, if the fundamental frequency is 1kHz, then the transient data from the last 1ms cycle would be used for the Fourier analysis.



Enabling the SPICE

Drawing the Schematic

Simulation dashboard panel

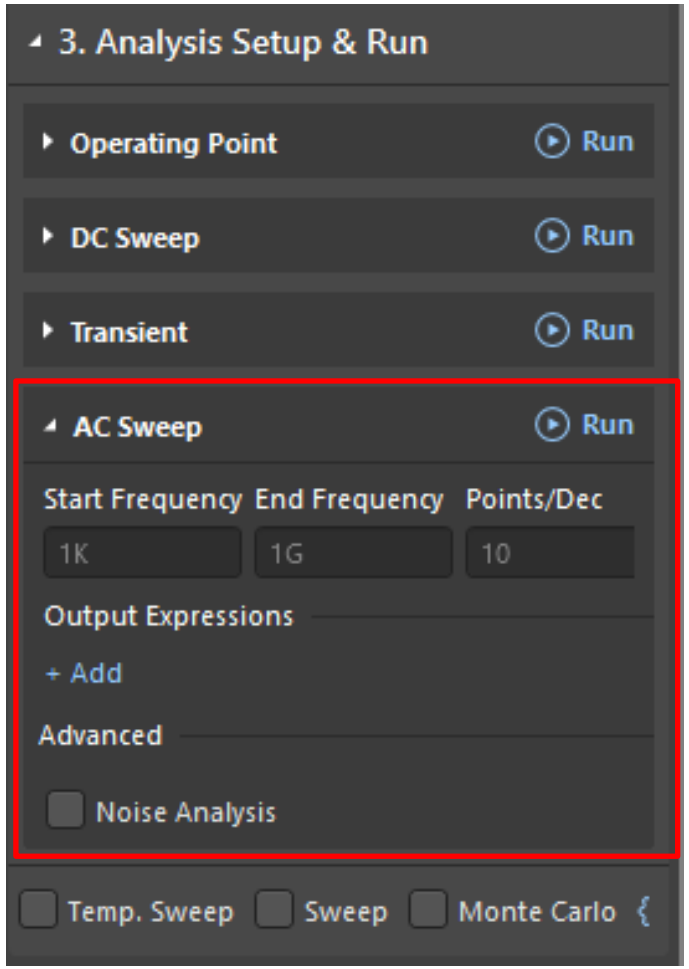
Exercise

Analysis Setup & Run

Quickly configure the required analysis type and the Output Expressions to be plotted, then run the simulation.

AC sweep - Linear or low-signal frequency response. Generates output that shows the frequency response of the circuit, calculating the small signal AC output variables as a function of frequency. It first performs an Operating Point analysis to determine the DC bias of the circuit, replacing the signal source with a fixed amplitude sine wave generator, then analyzing the circuit over the specified frequency range. The desired output of an AC Small Signal analysis is usually a transfer function (voltage gain, trans-impedance, etc).

- **Noise analysis** - measures the noise contributions of resistors and semiconductor devices by plotting the Noise Spectral Density, which is the noise measured in Volts squared per Hertz (V^2/Hz). Capacitors, inductors, and controlled sources are treated as noise free.



Enabling the SPICE

Drawing the Schematic

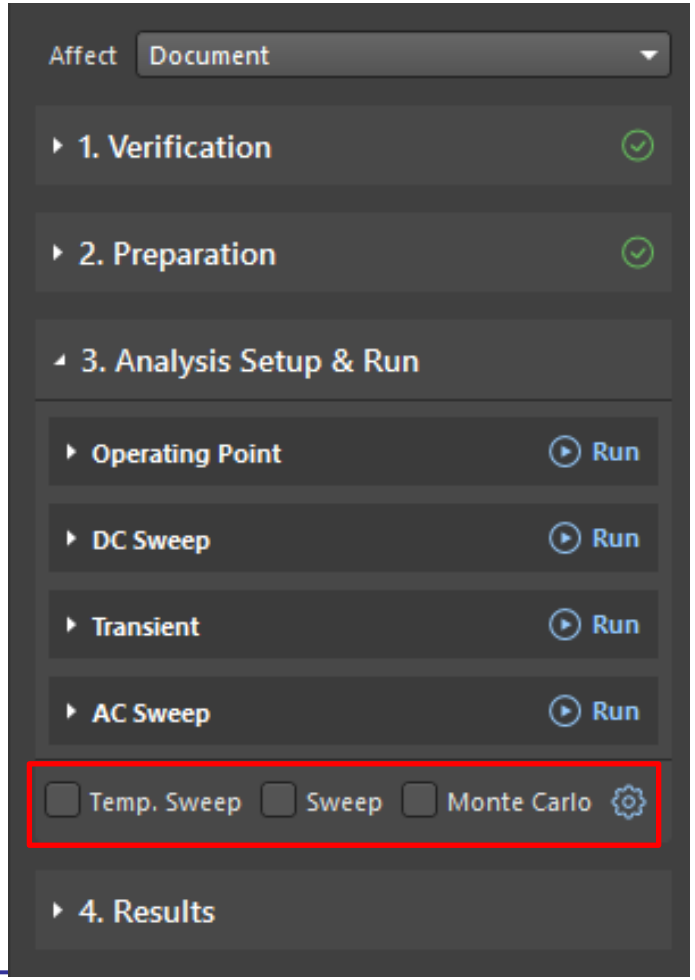
Simulation dashboard panel

Exercise

Analysis Setup & Run

Quickly configure the required analysis type and the Output Expressions to be plotted, then run the simulation.

- **Temperature sweep** - used to analyze the circuit at each temperature in a specified range, producing a series of curves, one for each temperature setting. The Simulator performs multiple passes of any of the standard analyses that are enabled (AC, DC Sweep, Operating Point, Transient, Transfer Function, Noise).
- **Parameters sweep** - used to sweep the value of a device in defined increments, over a specified range. The Simulator performs multiple passes of any of the standard analyses that are enabled (AC, DC Sweep, Operating Point, Transient, Transfer Function, Noise). The Parameter Sweep can vary basic components and models – sub-circuit data is not varied during the analysis. You can also define a Secondary parameter to be swept, when a Secondary parameter is defined the Primary parameter is swept for each value of the Secondary parameter.
- **Monte Carlo analysis** - used to perform multiple simulation runs with component values randomly varied across specified tolerances. The Simulator performs multiple passes of any of the standard analyses that are enabled (AC, DC Sweep, Operating Point, Transient, Transfer Function, Noise). The Monte Carlo analysis can vary basic components and models – sub-circuit data is not varied during the analysis.



Enabling the SPICE

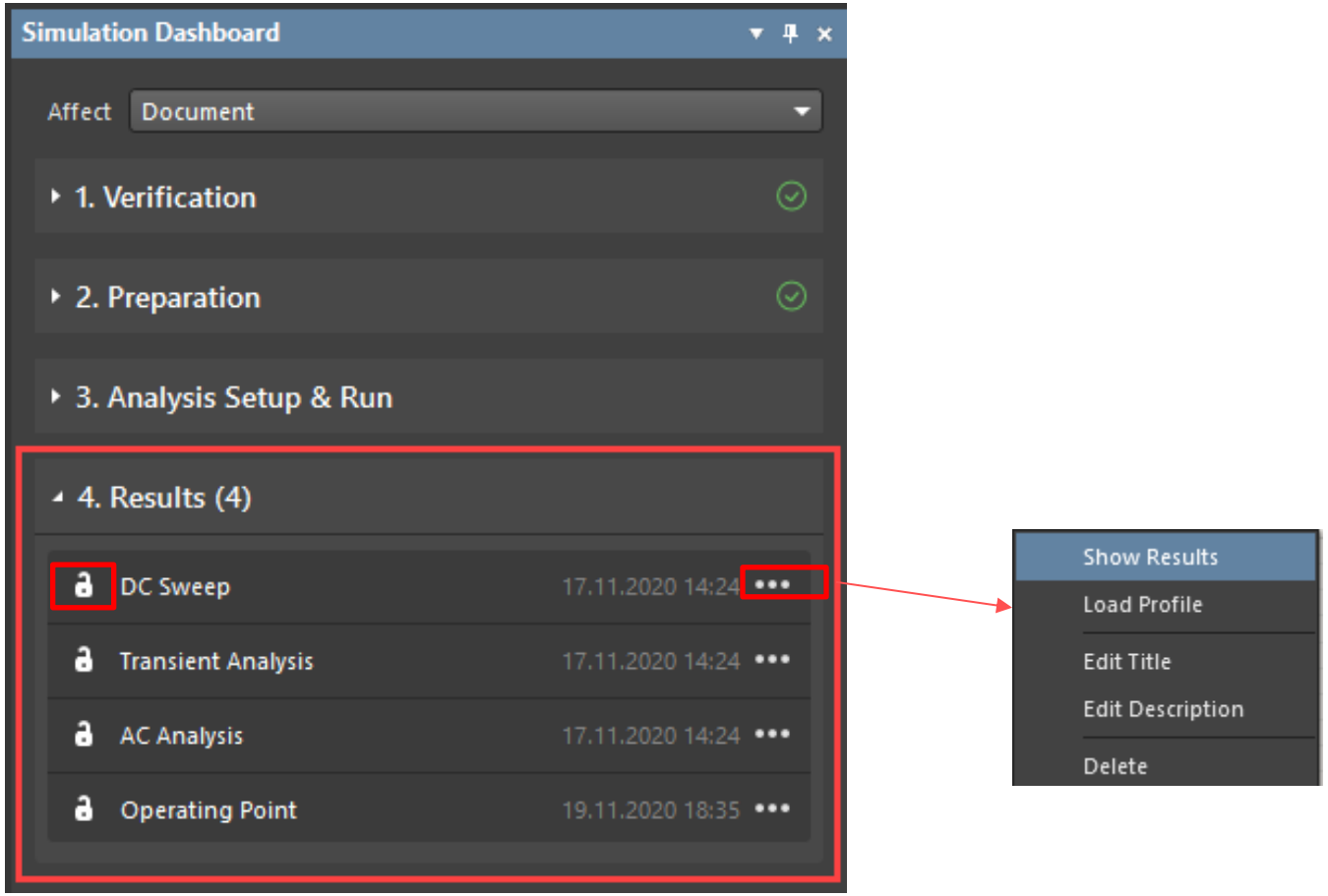
Drawing the Schematic

Simulation dashboard panel

Exercise

Results

Examine the previous analysis runs, double-click to re-open the waveform



Schematic Simulation - Exercises

Enabling the SPICE

Drawing the Schematic

Simulation dashboard panel

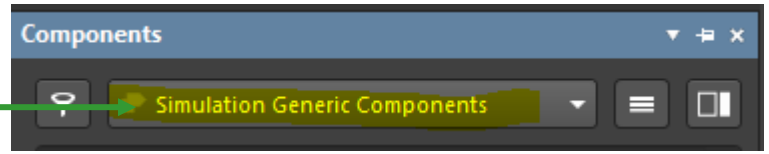
Exercise

Perform simulation to the circuits below:

1. **Simple voltage divider**
2. **Diode & Resistor circuit**
3. **Zener diode & Resistor circuit**
4. **555 Astable Multivibrator**
5. **Bandpass filter**

Guidance for each exercise:

1. Create a new PCB project
2. Creating a New Schematic Sheet
3. Adding components
4. Adding voltage source (VSRC)
5. Open Simulation dashboard panel
6. Perform Simulation



Enabling the SPICE

Drawing the Schematic

Simulation dashboard panel

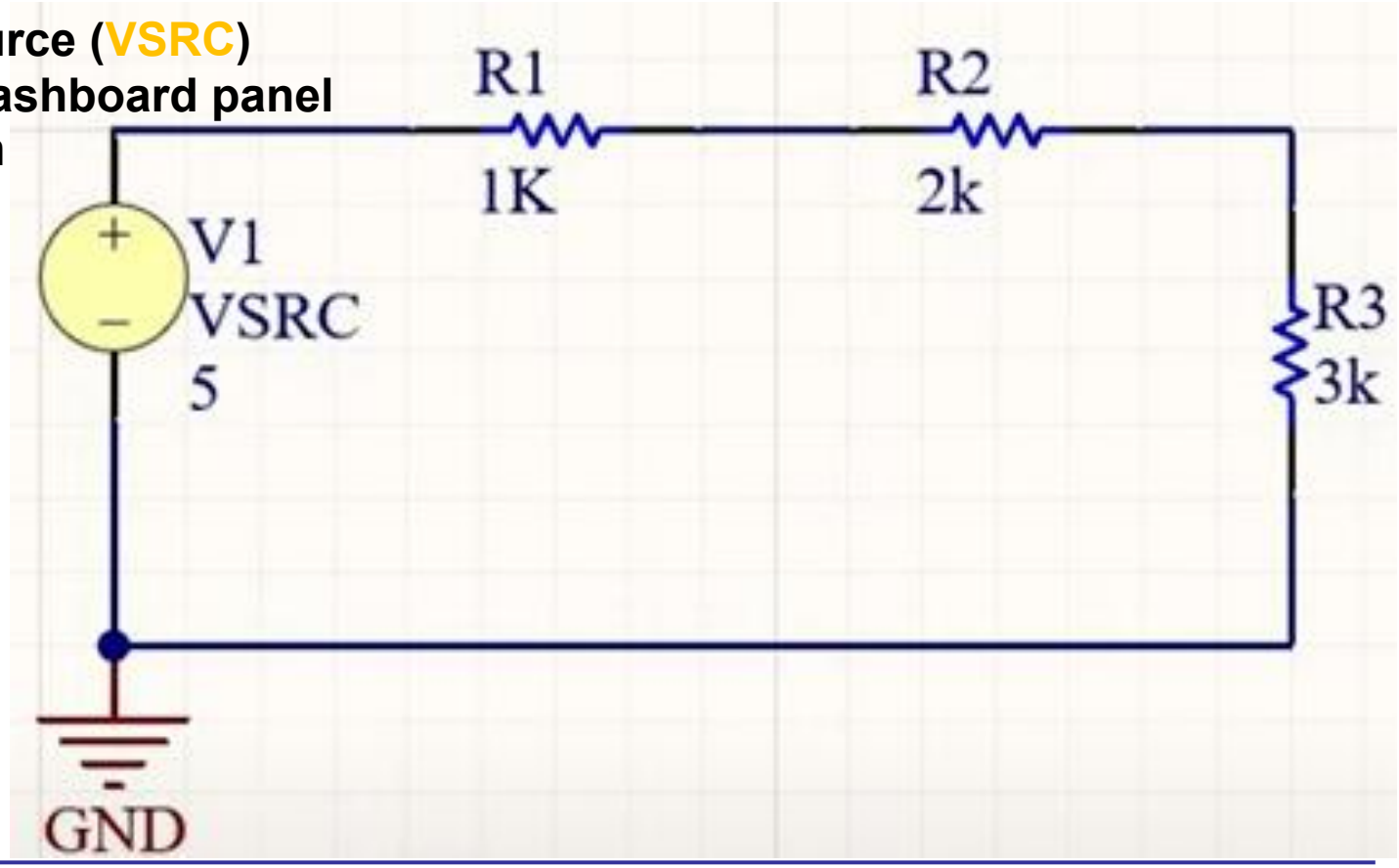
Exercise

1. Simple voltage divider

1. Create a new PCB project
2. Creating a New Schematic Sheet
3. Adding components
4. Adding voltage source (VSRC)
5. Open Simulation dashboard panel
6. Perform Simulation

a) Operating point –
VSRC DC source

b) DC Sweep –
VSRC DC source



Enabling the SPICE

Drawing the Schematic

Simulation dashboard panel

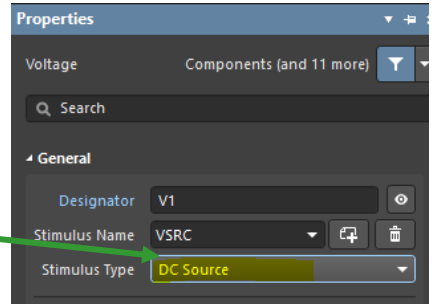
Exercise

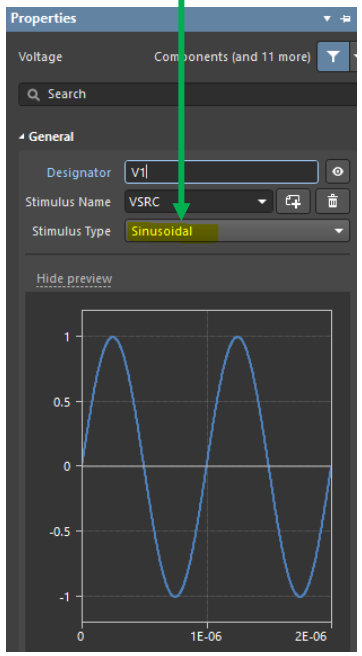
2. Diode Resistor circuit

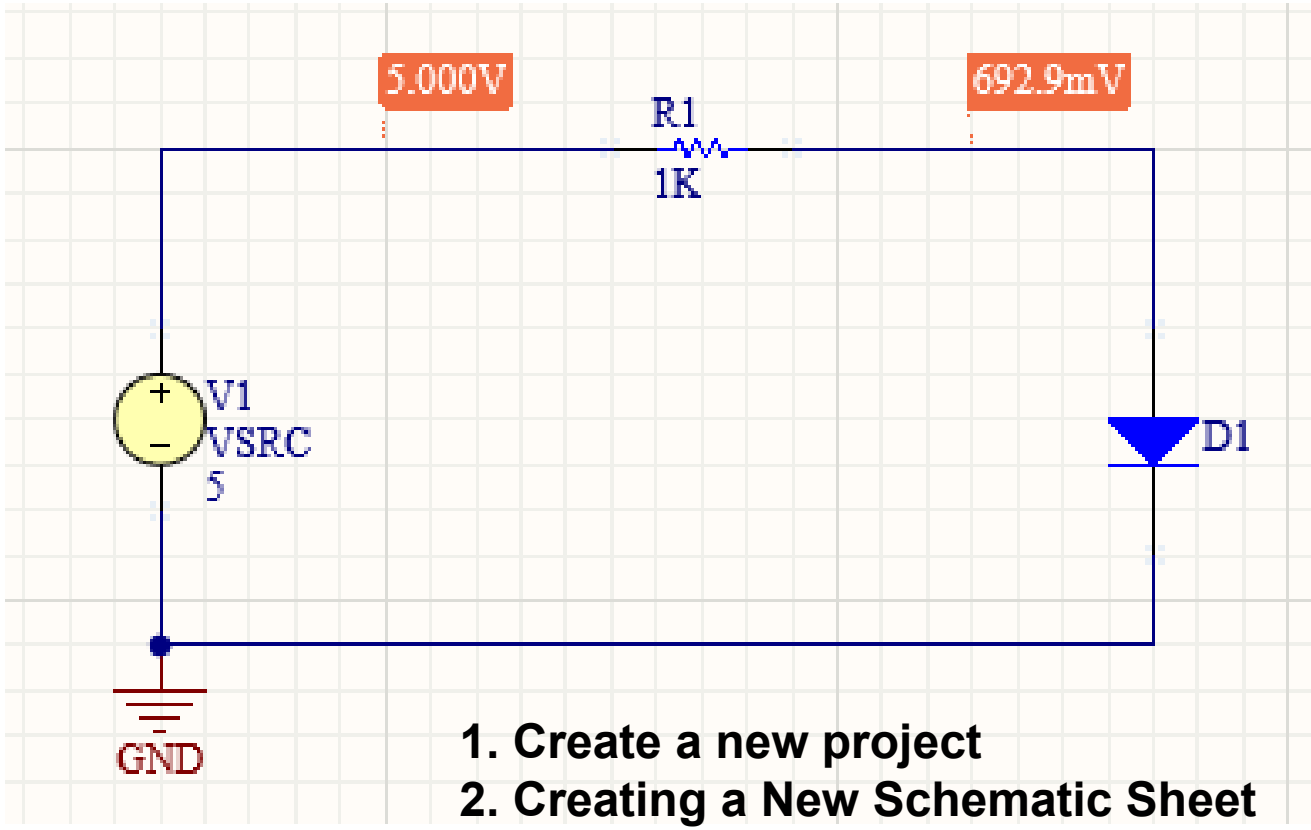
a) Operating point –
VSRC DC source

b) DC Sweep –
VSRC DC source

c) Transient –
VSRC sinusoidal







1. Create a new project
2. Creating a New Schematic Sheet

Enabling the SPICE

Drawing the Schematic

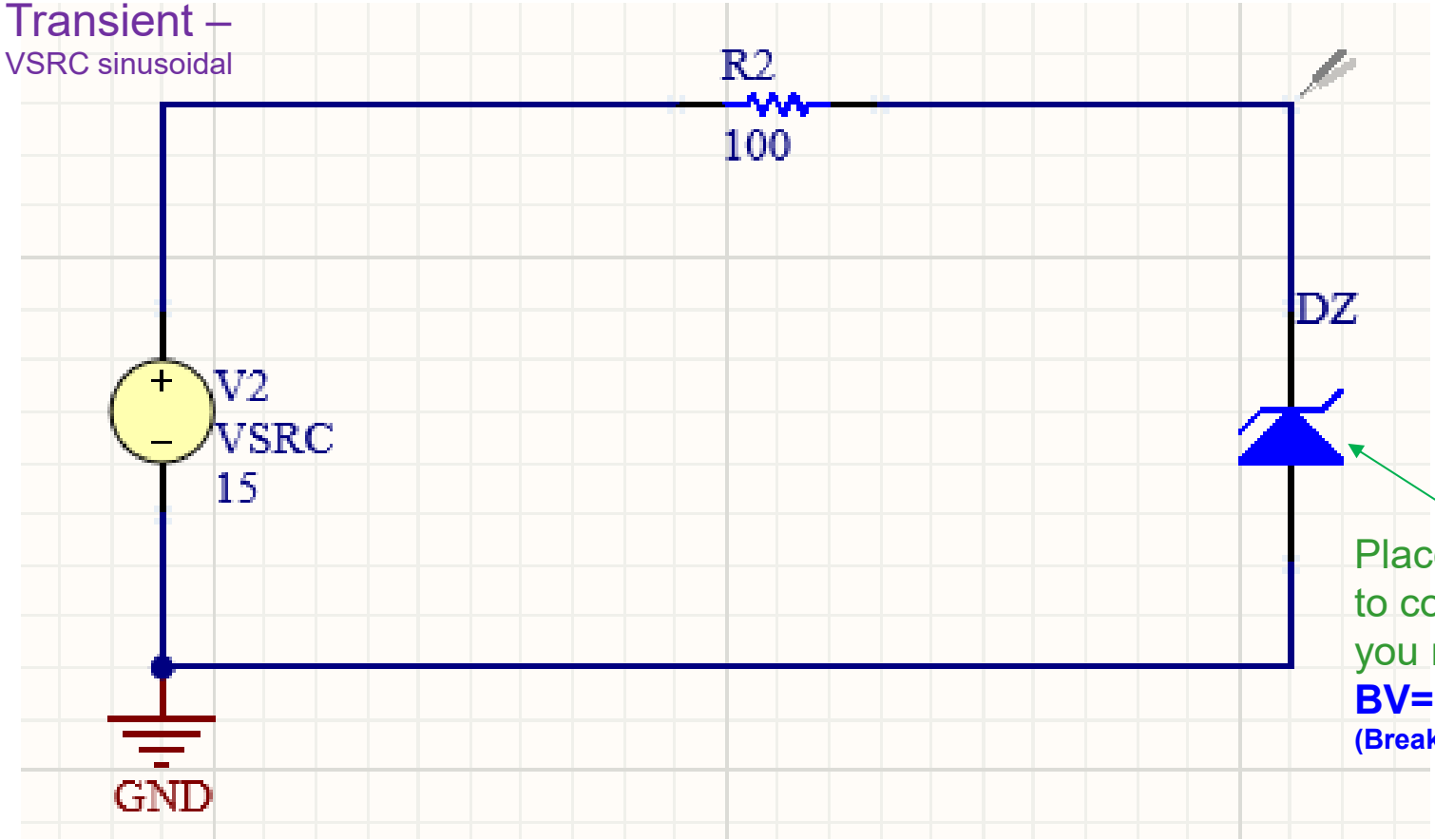
Simulation dashboard panel

Exercise

3. Zener diode Resistor circuit

a) DC Sweep –
VSRC DC source

b) Transient –
VSRC sinusoidal



Enabling the SPICE

Drawing the Schematic

Simulation dashboard panel

Exercise

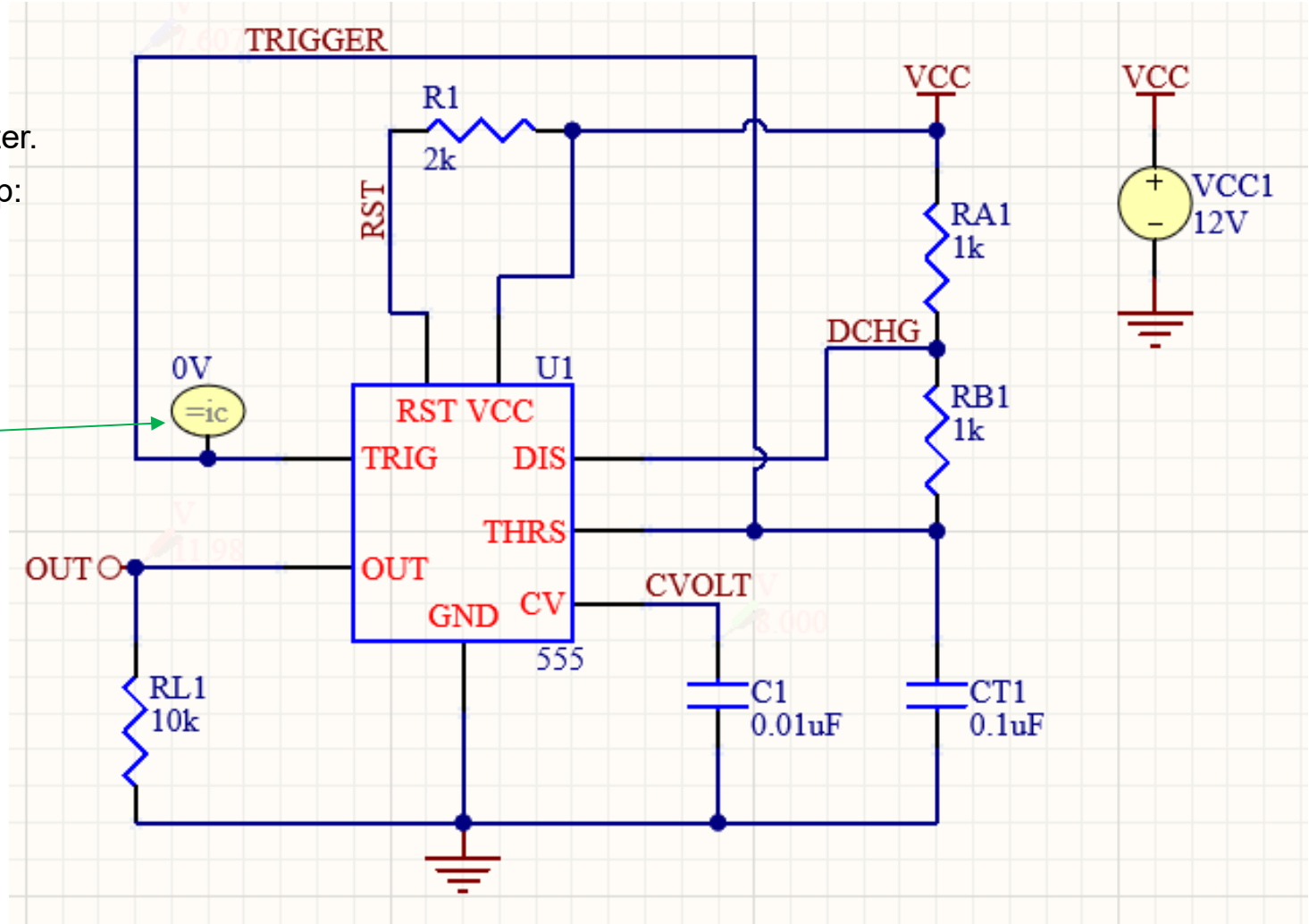
4. 555 Astable Multivibrator

a) Transient

1. Without Change parameter.
2. Change parameter sweep:
   CT1 / RA1 / RB1
   What's happened?

Initial Condition component

The OUT oscillation is around 1KHz



Schematic Simulation - Exercises

Enabling the SPICE

Drawing the Schematic

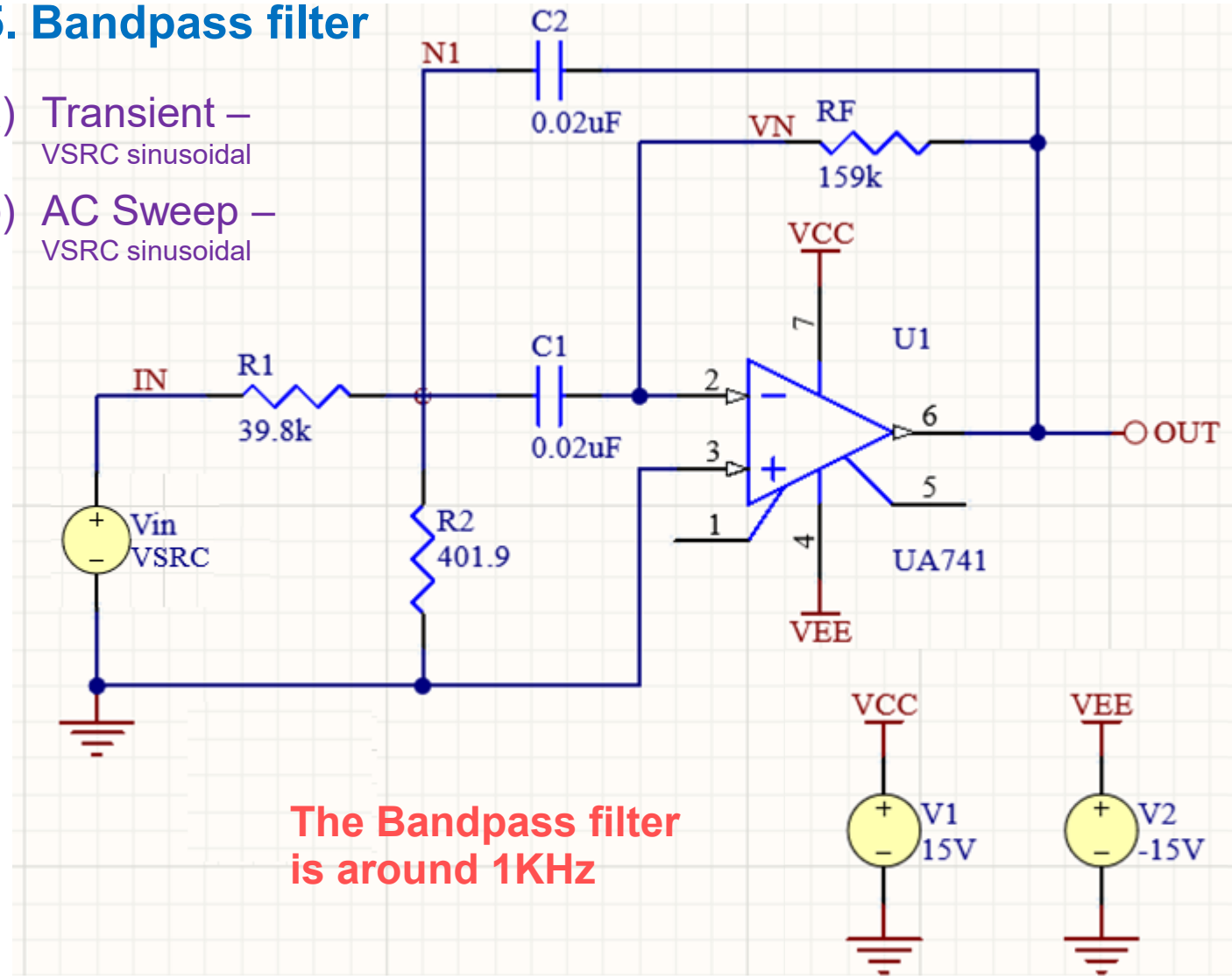
Simulation dashboard panel

Exercise

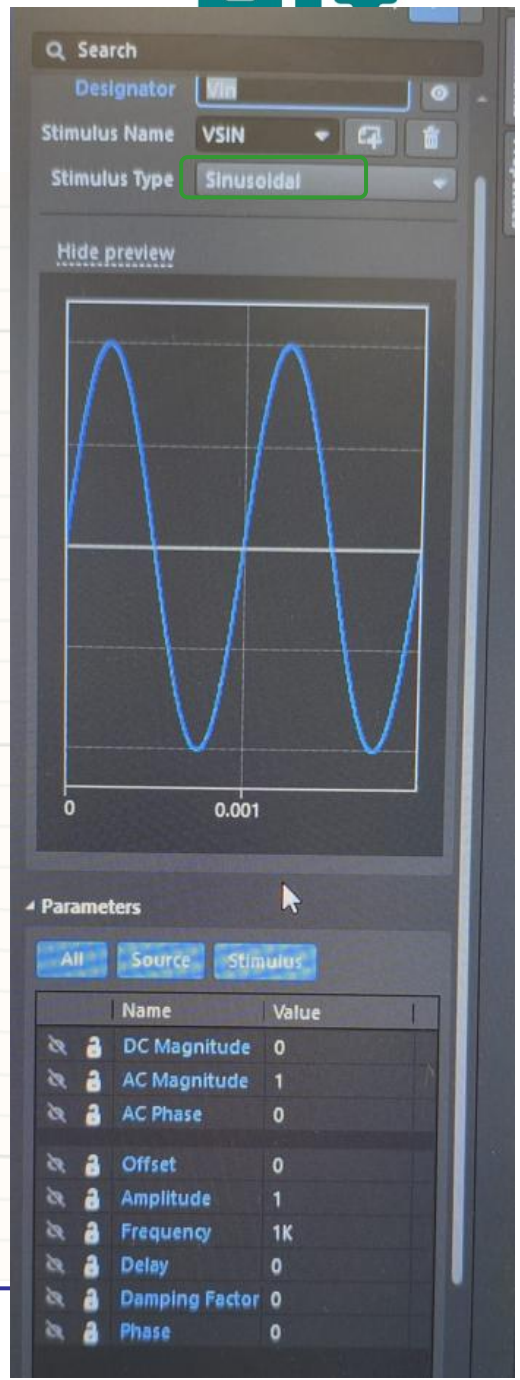
5. Bandpass filter

a) Transient –
VSRC sinusoidal

b) AC Sweep –
VSRC sinusoidal



The Bandpass filter is around 1KHz



Enabling the SPICE

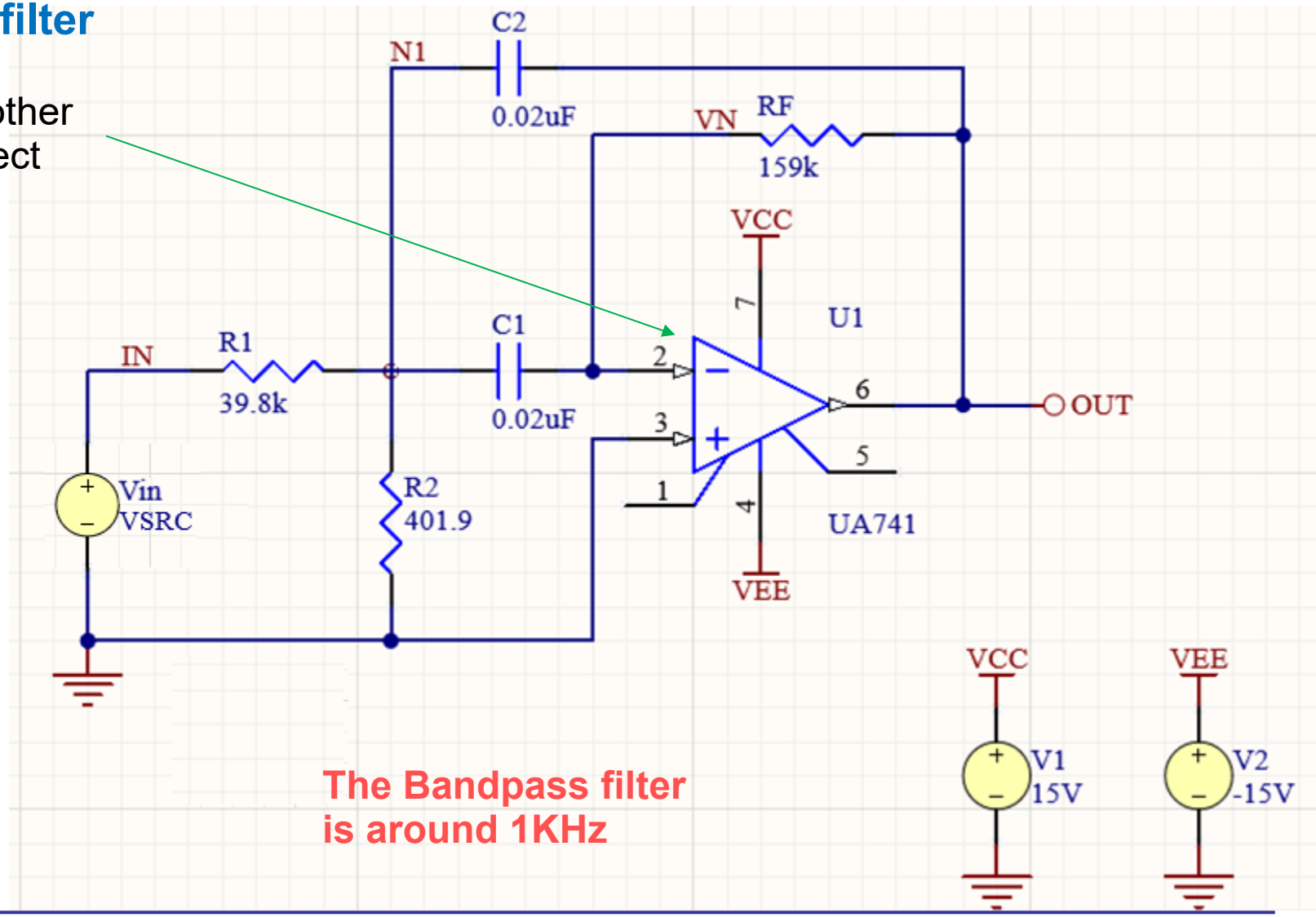
Drawing the Schematic

Simulation dashboard panel

Exercise

5. Bandpass filter

Import from another scheme or project



The Bandpass filter is around 1KHz

Enabling the SPICE

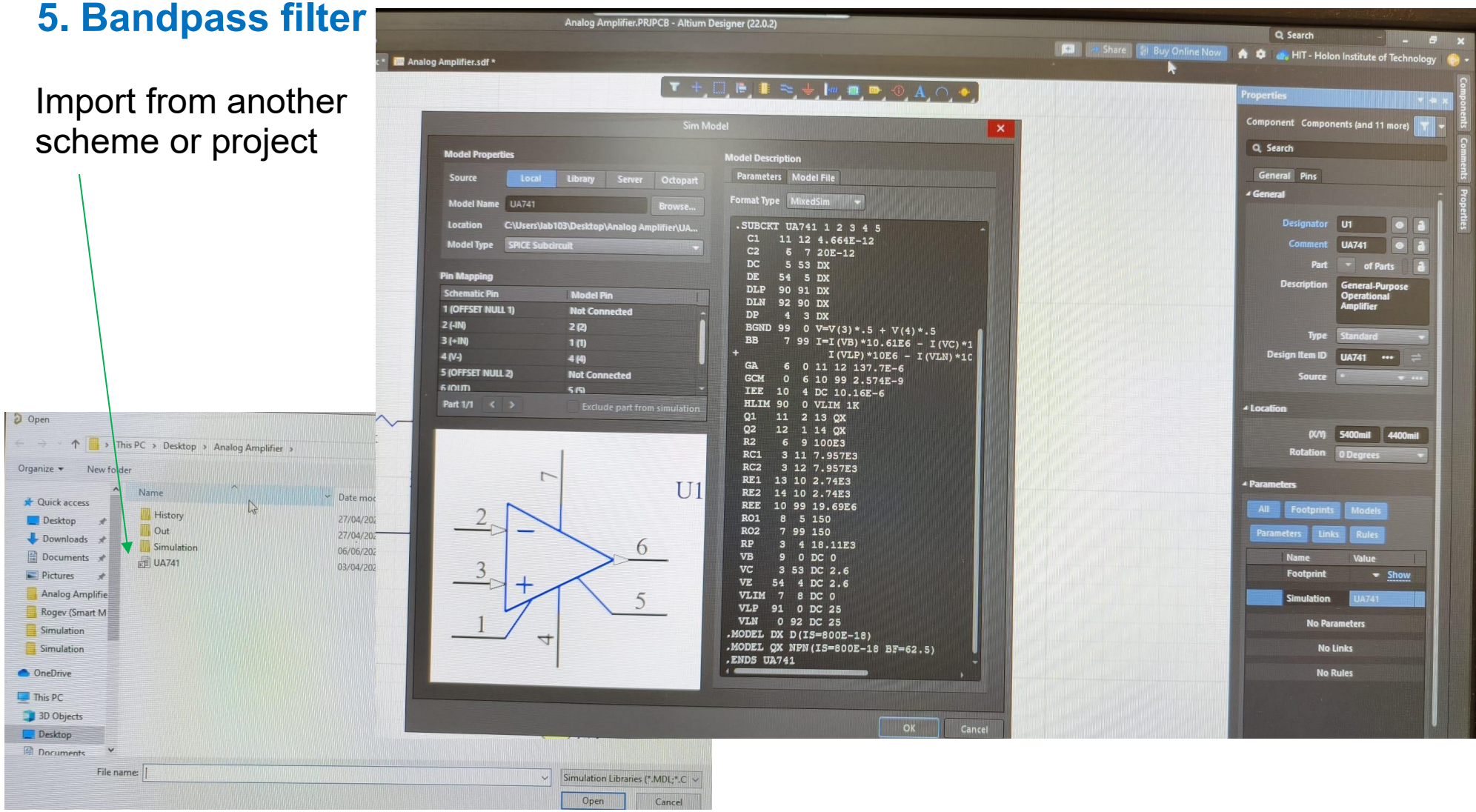
Drawing the Schematic

Simulation dashboard panel

Exercise

5. Bandpass filter

Import from another scheme or project



Enabling the SPICE

Drawing the Schematic

Simulation dashboard panel

Exercise

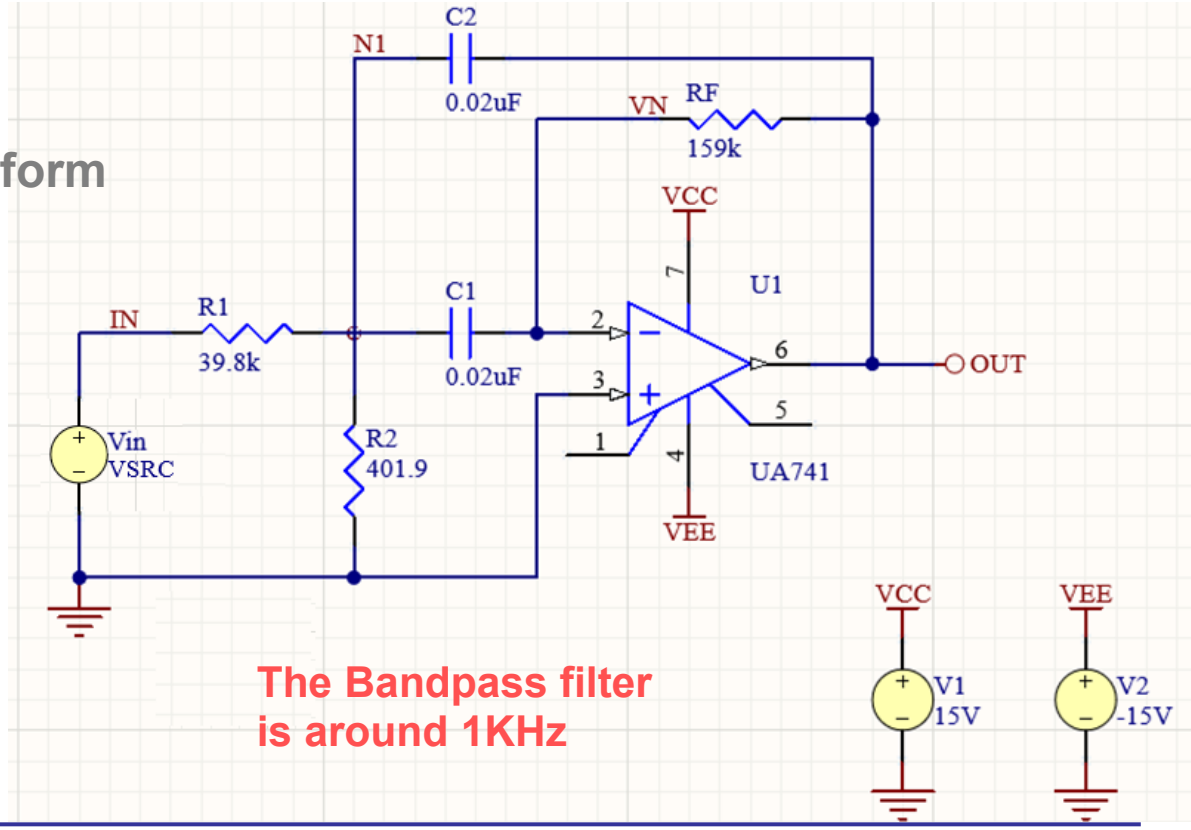
5. Bandpass filter

c) Create SCH library of components from the schematic:

Design » Make Schematic Library

d) Export plot data:

File » Export » Chart / Plot / Waveform



Enabling the SPICE

Drawing the Schematic

Simulation dashboard panel

Exercise

Links:

A Guide to SPICE Simulation in Altium Designer
https://www.altium.com/documentation/altium-designer/spice-simulation-guide

Working with the Simulation UI in Altium Designer
https://www.altium.com/documentation/altium-designer/simulation-user-interface